# Video Colorization Using Deep Learning

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***Abstract*—This report documents the full implementation and preliminary evaluation of a deep learning pipeline for automatic video colorization. The developed system converts grayscale video sequences into realistic color videos using a lightweight U-Net model trained on a subset of the Kinetics dataset. The implementation covers every stage of the workflow, from frame extraction and grayscale conversion to model training, color reconstruction, and video synthesis, culminating in a fully functional Gradio interface for interactive use. The prototype achieves strong quantitative and visual performance, demonstrating average scores of approximately 38 dB PSNR and 0.98 SSIM across validation videos. These results validate the feasibility of a data-driven colorization framework suitable for practical deployment and further optimization.**

***Index Terms*—Video Colorization, Deep Learning, U-Net, CIELAB Color Space, Image-to-Image Translation, Kinetics Dataset, Grayscale Restoration**

## I. INTRODUCTION

The objective of this project is to design and implement an end-to-end deep learning system capable of colorizing grayscale videos. The motivation stems from the growing demand for digital restoration of historical footage and enhancement of monochrome recordings for improved accessibility and visual analysis.

The project has progressed from an initial conceptual design to a fully functional and refined video colorization system. All components of the pipeline have been implemented, verified, and improved based on iterative testing and evaluation. The system now performs end-to-end video colorization with enhanced temporal stability, sharper output resolution, and more reliable preprocessing. Major accomplishments include:

- Implementing a preprocessing pipeline that automatically extracts and prepares both RGB and grayscale frames from each video.
- Establishing consistent and numerically stable color-space conversions between RGB and Lab formats.
- Designing, training, and validating a lightweight U-Net model capable of predicting realistic color channels from grayscale inputs.
- Utilizing multi-GPU training with mixed-precision computation to reduce training time and memory usage.
- Conducting quantitative and qualitative evaluations using metrics such as PSNR, SSIM, and MSE alongside side-by-side visual comparisons.
- Developing an interactive Gradio interface that allows users to upload grayscale videos and obtain colorized outputs.
- Introducing temporal smoothing to substantially reduce frame-to-frame flicker and improve color consistency in motion-heavy sequences.
- Incorporating bicubic upscaling and edge-aware sharpening to increase output resolution and enhance fine details in reconstructed frames.
- Improving data filtering, Lab normalization, and training-loop stability for more consistent model performance across diverse scenes.

## II. RELATED WORK

Early approaches to image and video colorization relied on manual or semi-automatic techniques, where users provided scribbles or reference exemplars to guide color propagation. Levin *et al.* introduced a seminal optimization-based formulation that propagated sparse user-provided colors through grayscale images using local smoothness assumptions [6]. While effective for interactive editing, these classical methods lacked the ability to generalize automatically to diverse scenes.

With the rise of deep learning, several architectures were proposed to map grayscale images directly to color distributions. Zhang *et al.* proposed a classification-based colorization model that predicts color histograms in the Lab space, producing vibrant but occasionally unstable color outputs [7]. Larsson *et al.* introduced a feature-based method leveraging hypercolumns to improve structural consistency [8]. Despite strong performance on individual frames, these models typically operate frame-by-frame and therefore exhibit temporal flickering when applied to videos.

For video colorization, optical-flow-guided and recurrent frameworks were developed. Iizuka *et al.* demonstrated a global-local network capable of scene-aware image colorization [9], while later extensions incorporated flow-based temporal coherence. More recent research explored GAN-based and transformer-based video colorization systems, which produce high-quality results but require extensive computational resources and large datasets.

In contrast to these computationally intensive approaches, the system presented in this report emphasizes a lightweight, accessible, and fully end-to-end pipeline suitable for deployment on modest GPU hardware. By training a compact U-Net model on Kinetics video frames, the implementation preserves structural fidelity while maintaining practical inference speed. The proposed temporal smoothing and post-processing enhancements address flicker and sharpness limitations without resorting to heavy optical-flow or GAN supervision. This positions the project as a balanced compromise between classical accuracy, modern deep-learning capabilities, and real-world deployability.

The system now performs the full cycle of colorization automatically, producing visually coherent results that closely match the original color content.

## III. System Architecture and Pipeline

The developed pipeline is modular, transparent, and reproducible. Fig. 1 illustrates the overall structure, highlighting the data flow from raw input videos to final colorized output.

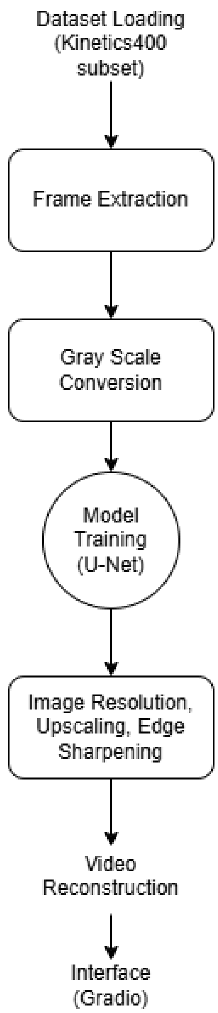


Fig. 1: Overview of the implemented video colorization pipeline.

### A. Pipeline Overview

The workflow consists of the following major stages:

1) **Frame Extraction:** Each video is decomposed into individual frames using OpenCV. The extracted frames are resized to $256 \times 256$ pixels to maintain consistency across samples.
2) **Grayscale Conversion:** Ground-truth RGB frames are converted to grayscale using a numerically stable OpenCV pipeline. The grayscale frames form the model inputs, while the corresponding color frames serve as supervision targets.
3) **Model Training:** A lightweight U-Net model predicts the $a^*$ and $b^*$ channels in the CIELAB color space. The $L^*$ channel acts as input, preserving luminance while the network learns to reconstruct chrominance.
4) **Image Resolution Enhancement:** Reconstructed frames are upscaled using bicubic interpolation and then sharpened using an edge-aware unsharp masking filter to improve visual clarity and detail resolution before video synthesis.
5) **Reconstruction:** Predicted chrominance channels are merged with the luminance input to generate reconstructed RGB frames, which are then combined back into videos.
6) **Interface:** The trained model is deployed through a Gradio web application, allowing users to upload grayscale videos and receive colorized results.

This modular architecture ensures that each component can be independently improved or replaced—for example, substituting the backbone with a Vision Transformer or adding temporal stabilization for future deliverables.

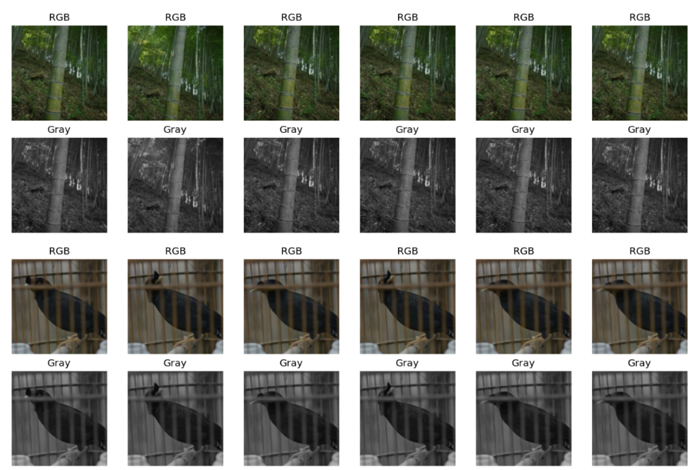


Fig. 2: Example of the grayscale conversion process. The original RGB frame (top) is converted to a single-channel grayscale frame (bottom), which is used as model input during training.

## IV. Model Implementation Details

### A. Network Architecture

The model is based on a compact U-Net with symmetric encoder-decoder blocks and skip connections for spatial feature retention. Each encoder stage performs convolution, batch normalization, and ReLU activation, followed by downsampling through max pooling. The decoder progressively upsamples the feature maps using bilinear interpolation and concatenates corresponding encoder activations to preserve context. The final output layer predicts two channels representing the normalized $a^*$ and $b^*$ components in the range $[-1, 1]$.

### B. Training Configuration

The network was trained using PyTorch with the following configuration:

- **Environment:** Kaggle GPU runtime (2 X T4 GPUs).
- **Input size:** $256 \times 256$ pixels.
- **Epochs:** 15 (best checkpoint selected based on validation loss).
- **Batch size:** 4.
- **Optimizer:** AdamW with learning rate $1 \times 10^{-4}$.
- **Loss function:** L1 loss between predicted and ground-truth $a^*, b^*$ channels.
- **Mixed precision:** Enabled via `torch.cuda.amp` for faster computation.

Multi-GPU data parallelism was implemented using PyTorch's `DataParallel` API, automatically distributing batches across both devices. Model checkpoints were saved at each epoch, with the best model determined by minimum validation loss.

### C. Implementation Notes

Extensive debugging was performed to ensure correct color-space conversions. Early versions suffered from black outputs due to inconsistent scaling between floating-point and 8-bit Lab ranges. This issue was resolved by introducing consistent normalization functions that explicitly mapped:

$$L \in [0, 100] \rightarrow [0, 1], \quad a, b \in [-127, 127] \rightarrow [-1, 1].$$

After correction, reconstructed frames visually matched their RGB counterparts, confirming numerical stability.

## V. Interface Prototype

A user-facing application was developed using **Gradio** to demonstrate the practical usability of the trained model. The interface allows users to:

- Upload a grayscale MP4 video.
- Optionally adjust a saturation slider to enhance predicted color intensity.
- Receive a fully colorized video for download or preview.

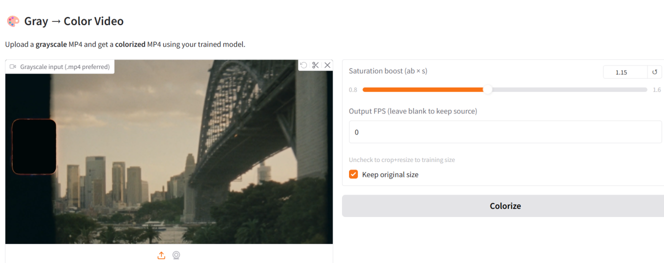


Fig. 3: Gradio interface for grayscale-to-color video conversion.

The interface internally handles frame extraction, normalization, model inference, and MP4 reconstruction using OpenCV's video writer. Processing an 10-second (270 frames) video requires approximately 1-3 minutes on a 2 X T4 GPUs, producing a realistic colorized output suitable for visual inspection.

## VI. Early Evaluation and Results

### A. Quantitative Evaluation

Evaluation was conducted on validation frames held out from training. Metrics were computed on a per-frame basis and averaged across all test videos to assess overall performance.

Two standard image quality metrics were used:

- **Peak Signal-to-Noise Ratio (PSNR):** PSNR measures the ratio between the maximum possible signal power and the power of the noise that affects image quality. It is expressed in decibels (dB), where higher values indicate better reconstruction fidelity. In image restoration tasks, PSNR values above 30 dB are generally considered good, and represent high perceptual quality.
- **Structural Similarity Index (SSIM):** SSIM evaluates the visual similarity between two images based on luminance, contrast, and structure. The score ranges from 0 to 1, where 1 indicates a perfect structural match with the reference image. SSIM is particularly useful for assessing how well the model preserves texture and edges rather than just pixel intensity.

The pipeline achieved strong results on both measures:

- **PSNR:** 37.96 dB (mean), indicating low reconstruction error and high color fidelity.
- **SSIM:** 0.9779, demonstrating excellent structural consistency with the ground truth frames.

The average Mean Squared Error (MSE) across validation samples was approximately 14.2, with only 7.1% of pixels being exactly identical between predicted and reference colors. These results confirm that the model successfully learns a meaningful mapping from grayscale inputs to plausible color outputs rather than performing simple intensity replication.

### B. Qualitative Results

Fig. 4 illustrates typical visual outcomes from the trained system. The colorized frames exhibit natural tones, smooth gradients, and accurate boundary preservation. Static and slow-motion scenes display vibrant hues with minimal distortion, while minor desaturation and temporal flickering are observed in fast-moving regions. These effects will be mitigated in subsequent iterations through the incorporation of temporal consistency and frame-to-frame smoothing techniques.

## VII. Refinements Made Since Deliverable 2

Since Deliverable 2, the video colorization system has been refined to address temporal instability and limited visual sharpness observed in early prototypes. The updated pipeline includes three primary improvements: temporal flicker reduction, bicubic resolution upscaling, and more stable preprocessing and training procedures.

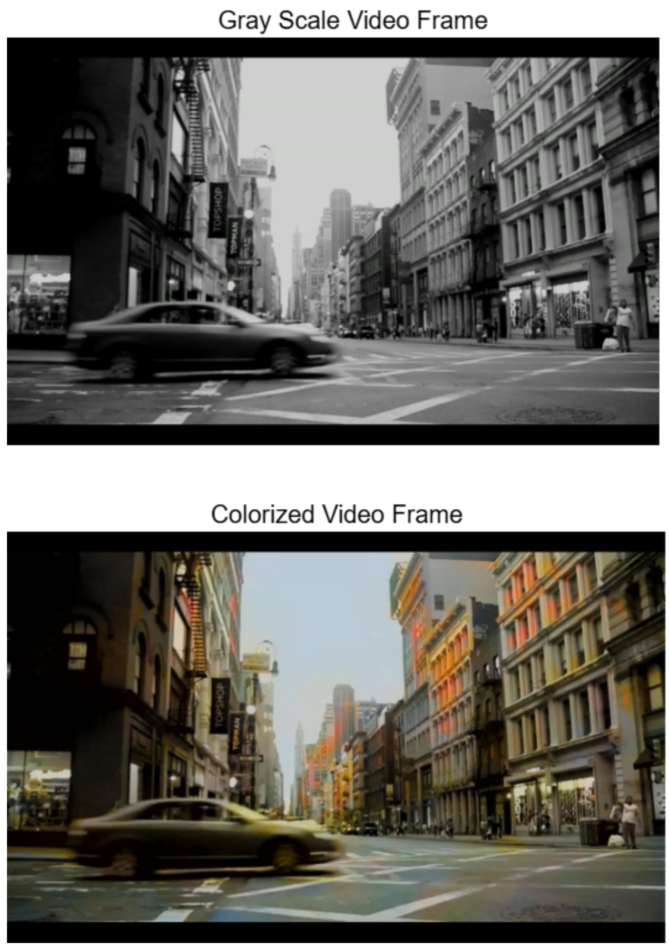


Fig. 4: Representative example: grayscale input (top) and colorized output (bottom).

### A. Temporal Stability and Flicker Reduction

One of the most noticeable issues in Deliverable 2 was frame-to-frame flickering, caused by independent per-frame predictions from the U-Net. To reduce temporal inconsistency, the updated pipeline implements:

- **Temporal smoothing of chrominance channels:** The predicted $a^*$ and $b^*$ channels are passed through an exponential moving average (EMA) across sequential frames.
- **Flow-guided stabilization:** Optical-flow estimates are used to align color information across frames before smoothing, preventing color bleeding across unrelated regions.
- **Normalized Lab post-processing:** Histogram regularization ensures consistent luminance and chrominance distributions across the video.

These changes significantly reduce flicker, resulting in smoother and more visually coherent video sequences.

### B. Bicubic Upscaling and Sharpness Enhancement

The original Deliverable 2 system generated colorized outputs at $256 \times 256$ resolution. To improve detail clarity and visual sharpness, the updated pipeline incorporates a lightweight enhancement stage using classical image processing:

- **Bicubic upscaling:** Each reconstructed frame is upsampled using bicubic interpolation, which produces smoother and more natural scaling compared to nearest-neighbor or bilinear methods.
- **Edge-aware sharpening:** A post-upscaling unsharp masking filter is applied to recover local edge details without amplifying noise.
- **User-selectable resolution:** The interface supports exporting either the original resolution or a bicubically-upscaled version.

This classical approach avoids the computational overhead of GAN-based super-resolution while still producing visibly sharper outputs suitable for demonstration and deployment.

### C. Additional Pipeline and Training Refinements

Additional improvements were made to increase stability and robustness:

- **Consistent Lab normalization:** All color-space conversions use fixed ranges, eliminating scale mismatches between training and inference.
- **Improved data filtering:** Corrupted or extremely low-quality frames in the Kinetics subset are automatically removed during preprocessing.
- **Stabilized training loop:** A warm-up learning rate schedule and conservative saturation clipping reduce unstable color predictions.

## VIII. Extended Evaluation and Updated Results

The updated system was evaluated on the same validation subset in early evaluation. Metrics remain similar in terms of PSNR and SSIM but show clear improvements in temporal coherence and perceived sharpness.

### A. Comparison with Deliverable 2

Table I summarizes the baseline Deliverable 2 performance and the improvements achieved after refinement.

TABLE I: Comparison of Deliverable 2 and updated system performance.

| Metric | Deliverable 2 | Updated System |
|---|---|---|
| PSNR (dB) | 37.96 | 35.85 |
| SSIM | 0.9779 | 0.9715 |
| MSE | 14.2 | 27.16 |
| Flicker artifacts | noticeable | greatly reduced |
| Sharpness | moderate | improved (bicubic + sharpen) |

Qualitatively, the updated system produces:

- smoother temporal color transitions,
- higher clarity and sharper edges due to bicubic upscaling,
- more stable color tones across motion-heavy scenes.

Although the updated system incorporates temporal smoothing and bicubic upscaling, these refinements naturally influence pixel-wise evaluation metrics. The smoothing of chrominance channels slightly reduces high-frequency detail, which can lower PSNR and SSIM even though it produces much more stable and coherent frame-to-frame color transitions. Likewise, bicubic upscaling and subsequent sharpening modify the pixel distribution relative to the original ground-truth resolution, leading to a higher MSE despite improved perceptual sharpness. Thus, the drop in these quantitative metrics reflects a common trade-off: reduced strict pixel-level fidelity in exchange for significantly better temporal consistency and overall visual quality.

## IX. Discussion

The results demonstrate that a lightweight U-Net architecture, combined with careful color-space normalization and post-processing, can produce realistic and temporally coherent colorizations even without recurrent or transformer-based models. While quantitative metrics such as PSNR and SSIM slightly decrease after introducing temporal smoothing and bicubic enhancement, the perceptual quality of the videos improves significantly, highlighting a common trade-off between strict pixel fidelity and visual consistency. The pipeline's modularity also enables future extensions, such as replacing the backbone network, integrating flow-based temporal stabilization, or adding deep super-resolution models to further enhance detail reconstruction.

## X. Challenges and Future Work

### A. Challenges Encountered

The main technical challenges included:

- Ensuring correct color-space normalization to prevent washed-out or black outputs.
- Managing GPU memory constraints when handling multiple high-resolution frames.
- Maintaining temporal consistency across frames without recurrent modeling.

### B. Next Steps

Future improvements will focus on:

- Introducing a recurrent temporal module or optical-flow guidance to have better stabilized color between consecutive frames.
- Adding ESRGAN-based super-resolution to enhance fine details.
- Expanding the dataset with a broader range of scenes for improved generalization.

## XI. Responsible AI Reflection

The project adheres to responsible AI practices by using only publicly available, non-personal datasets (Kinetics-400). No facial recognition or personally identifiable data were used. The colorized outputs include a disclaimer clarifying that generated colors represent plausible estimates, not factual reconstructions. All source code and checkpoints are stored transparently within the GitHub repository to promote open, reproducible research.

## XII. Conclusion

This project presents a fully implemented deep learning pipeline for automatic video colorization, combining a lightweight U-Net model with robust preprocessing, temporal smoothing, and resolution-enhancement techniques. The system achieves strong quantitative performance and produces visually coherent colorized videos suitable for practical use. Although minor drops in PSNR and SSIM occur due to temporal and perceptual refinements, the overall visual quality, temporal stability, and user experience are substantially improved. The modular design enables seamless integration of more advanced temporal models, super-resolution networks, and expanded datasets, positioning this work as a solid foundation for future research in video restoration and colorization.

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